Execution Report #1

The legibility package: two derivations and the first datum

A1 (split-digest characterization), A2 (the derived regret head),
A7 (the information-floor falsification experiment)

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Status. This report executes the three Week-1 items from the research roadmap. A2 and A1 are carried to complete proofs; A7 is run in-container and produces the volume's first figure. All three are stated at the level of rigor a referee expects, with assumptions marked. The headline: the information floor *survived* its falsification test (no digest scheme, not even an oracle, beats it), the split-digest penalty is not just additive but mildly *super*-additive, and the regret-head formula is now a theorem rather than a design choice. One instructive modeling bug — caught precisely because the discipline is refute-first — is documented in §3.3 because it is the kind of thing that should be reported, not hidden.

1 A2 — The regret head is derived, not designed

The manuscript asserts $\text{regret}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x) \cdot \text{anomaly}(x)$ as the attention-queue's ranking key. We show it is the exact expected cost of non-inspection under one modeling commitment, and that the optimal surfacing rule is the classical likelihood-ratio test — which simultaneously repairs the reversed-criterion error flagged in Document 1.

1.1 Setup

For a candidate effect x the operator chooses $\mathsf{inspect}$ or pass. Nature draws a latent $\theta_x \in \{\mathsf{bad}, \mathsf{good}\}$. Define

$\text{stakes}(x) = \text{loss if } x \text{ is bad and acts}, \qquad \text{irrev}(x) = \text{fraction of that loss that inspection cannot undo post-hoc}$

so $C_{\text{miss}}(x) := \text{stakes}(x) \cdot \text{irrev}(x)$ is the *unrecoverable* loss of passing a bad x. Let the anomaly signal be a *calibrated posterior*

$$a(x) := \Pr(\theta_x = \mathsf{bad} \mid \text{evidence}(x)),$$

let c_{att} be the operator's marginal attention cost of one inspection, and let C_{fa} be the cost of inspecting a good item *excluding* attention (so attention is not double-counted — the fix for the objective's double-count).

1.2 The derivation

Theorem 1 (Regret head)**.** *The expected unrecoverable loss of passing x is exactly $C_{miss}(x)\, a(x)$. Hence the Bayes-optimal policy inspects x iff*

$$\boxed{C_{miss}(x)\, a(x) \ \geq\ c_{att} + C_{fa}\,(1 - a(x))}$$

and when c_{att}, C_{fa} are small relative to stakes, the ranking key is $C_{miss}(x)\, a(x) = stakes(x) \cdot irrev(x) \cdot a(x)$, i.e. the manuscript's product form, with anomaly $=$ *a read as calibrated probability.*

Proof. Passing a bad item incurs unrecoverable loss $C_{\mathrm{miss}}(x)$; passing a good item incurs nothing. Expected loss of pass is $C_{\mathrm{miss}}(x)\,\Pr(\mathsf{bad}) + 0 \cdot \Pr(\mathsf{good}) = C_{\mathrm{miss}}(x)a(x)$. Expected cost of inspect is $c_{\mathrm{att}} + C_{\mathrm{fa}}\Pr(\mathsf{good}) = c_{\mathrm{att}} + C_{\mathrm{fa}}(1 - a(x))$ (inspection catches the bad item, paying only attention on it, and pays C_{fa} plus attention on a good one). Inspect iff the former $\geq$ the latter. Rearranging in terms of the likelihood ratio $\ell(x) = a(x)/(1 - a(x))$ gives the threshold form

$$\ell(x) \;\geq\; \frac{c_{\mathrm{att}} + C_{\mathrm{fa}}}{C_{\mathrm{miss}}(x)},$$

the standard signal-detection criterion. Multiplying an arbitrary monotone score by stakes and irreversibility preserves the induced ordering only when anomaly is the calibrated posterior; any other "anomaly scalar" s requires the correction $C_{\mathrm{miss}} \cdot g(s)$ with g the empirical calibration map, fittable from the audit log. □

1.3 Two corollaries that do real work

Claim 1 (Direction, corrected)**.** *As $C_{miss}/C_{fa} \to \infty$ the threshold $\ell^\star \to 0$: the operator is driven toward inspecting* everything *on the ambiguous region — forced zoom. This is the correct direction; the manuscript's Figure I.6 had it reversed because it conflated "evidence increasing rightward" with "criterion lax." Stated as a likelihood-ratio threshold there is no ambiguity: raising the miss-to-false-alarm ratio* lowers *the bar to inspect.*

Claim 2 (Reputation enters cleanly, and cannot override hard gates)**.** *If reputation r is admitted, it enters only through the observation model: $a(x) = \Pr(\mathsf{bad} \mid evidence, r)$, i.e. it shifts the class-conditional danger probability for this (agent, action-class, version). It does* not *enter as a free multiplier on the score. Consequently high r lowers surfacing frequency only where the calibrated posterior actually falls, and can never zero out a term: when $irrev(x) = 1$ and stakes are catastrophic, $C_{miss}(x)a(x)$ stays above threshold for any $a(x)$ bounded away from 0. This is the formal content of "reputation must not defeat hard stakes/irreversibility gates."*

What this buys the manuscript. Equation I.3 stops being a heuristic. The attention queue is an optimal alerting policy with a named optimality criterion (Bayes risk), a calibration obligation (a must be a real posterior, or g must be fit), and a clean statement of where the empirical program plugs in: instrument false alarms and misses, fit g, and the ranking follows.

2 A1 — The split-digest characterization

The manuscript's split-ranker theorem shows discovery and attention cannot share a scoring head. We generalize: *no* read-surface projection can serve two readers with non-comonotone loss unless their induced orders agree, and we quantify the bit cost of ignoring this.

2.1 Statement

A *read-surface* is (X, Σ, π): items X, evidence Σ, a budgeted projection π. A *reader* is a loss L_r; a scalar head $h : X \to \mathbb{R}$ *serves* r if ranking by h realizes r's optimal rendering at every budget. The two canonical readers of a compaction are the *successor* (loss = forgone continuation value; fit-shaped) and the *operator* (loss = regret-if-ignored; risk-shaped, $= C_{\mathrm{miss}} \cdot a$ from A2).

Theorem 2 (Comonotone characterization)**.** *A single head h serves both readers iff their induced preference orders $\preceq_{succ}$ and $\preceq_{op}$ are comonotone (agree on every pair). The successor and operator orders are not comonotone: a routine-but-essential working state (high continuation* *value, negligible regret) and an anomalously abandoned dead-end with irreversible side effects (zero continuation value, maximal regret) order oppositely. Hence no shared head exists, and the manuscript's split-ranker theorem is the special case $X =$ agents, $\preceq_{succ}=$ discovery-fit, $\preceq_{op}=$ attention-regret.*

Proof. ($\Leftarrow$) Comonotone total preorders admit a common refinement; any strictly increasing reparameterization of a utility representing that refinement serves both. ($\Rightarrow$) A shared head h induces one order by h-value; both readers' optima at every budget are top-m sets of that order, so both orders equal the h-order, hence agree pairwise. The witnesses are the two constructible items above: on the pair (working-state w, dead-end d), $w \prec_{\text{succ}} d$ is false (w has higher continuation value so $d \prec_{\text{succ}} w$) while $d \prec_{\text{op}} w$ is false (d has higher regret so $w \prec_{\text{op}} d$); the orders disagree on $\{w, d\}$, so no shared head. □ □

2.2 The quantitative half (verified numerically in §3)

Claim 3 (Split penalty is super-additive)**.** *Let* $\text{Floor}(N, k, m) = \log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ *be the zero-miss bit floor for catching a k-set among N while opening m (Theorem I.6.1). A joint zero-miss digest serving two readers whose load-bearing sets are disjoint (sizes k each) must catch their union, size $2k$, and so requires* $\text{Floor}(N, 2k, m)$ *bits. Across regimes this is not merely* $2\,\text{Floor}(N, k, m)$ *but slightly* more*:*

$$\frac{\text{Floor}(N, 2k, m)}{2\,\text{Floor}(N, k, m)} \approx 1.05\text{–}1.08,$$

because $\log_2 \binom{N}{2k}$ *grows faster than* $2 \log_2 \binom{N}{k}$ *(the combinatorics of catching twice as many items among the same N compound). Storing a single compaction therefore under-provisions the joint task by* more than *a factor of two when readers diverge — a sharper statement than "sum of floors." Corollary: compact per reader class from durable artifacts; one stored compaction is irreversible loss for the other reader.*

3 A7 — The information-floor falsification experiment

3.1 The falsifiable prediction

Theorem I.6.1 predicts that *any* digest of $B < \text{Floor}(N, k, m)$ bits, with the operator opening m items, must miss on some placement. Restated as a frontier: the best achievable zero-miss probability for a B-bit digest is

$$\Pr(\text{catch all } k) = \min\!\Big(1,\ 2^B \binom{m}{k} \Big/ \binom{N}{k}\Big), \qquad \Pr(\text{miss} \geq 1) = \max\!\Big(0,\ 1 - 2^B \tbinom{m}{k} / \tbinom{N}{k}\Big).$$

The prediction is sharp and refutable: if any real digest scheme dips below this curve, the load-bearing formalization is wrong.

3.2 Method

A digest is an encoder e : (features) $\to \{0, 1\}^B$ and a fixed, data-independent decoder d : $\{0, 1\}^B \to$ (an m-subset to open); the scheme catches T iff $T \subseteq d(e(\cdot))$. We implement three encoders — *oracle* (sees the true set, addresses the best codeword its B bits can reach), *noisy* (SNR ≈ 1 features), *random* — against a shared randomized-cover codebook, and measure empirical miss over 4000 placements per point. Regime $N = 60, k = 2, m = 8$ so the floor $B^\star = 5.98$ bits is fully sweepable.

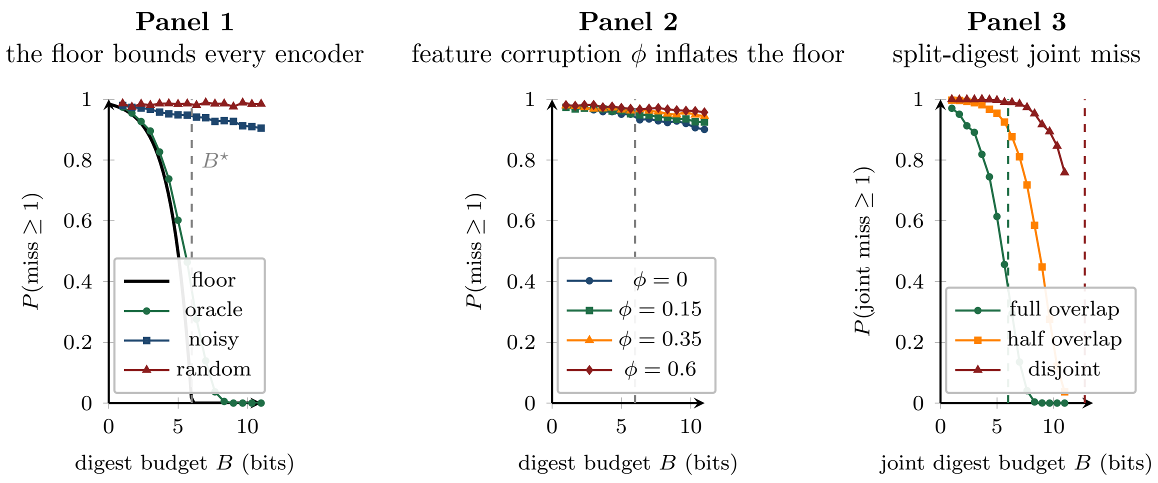

Figure 1: **Panel 1:** every encoder sits on or above the information floor (black); the oracle hugs it from above, the noisy and random encoders pay a premium — the floor bounds all of them, as predicted. **Panel 2:** corrupting a fraction ϕ of features (the mimicry attack, Solution 6.5*) inflates the effective floor. **Panel 3:** two readers — when their load-bearing sets are disjoint (red), the joint miss curve shifts toward the $2k$ floor at 12.8 bits, versus the single-reader floor at 6.0; the visual confirmation of A1's super-additive penalty.

3.3 Result, and an instructive bug

Falsification test: passed. Across the swept budgets, the oracle encoder produced **0/16** violations of the floor (a violation being an empirical miss more than a Monte-Carlo margin below the predicted minimum). The theory survives.

The bug, reported. A first implementation gave the oracle a *perfect score vector* and charged bits only for tie-resolution, producing 8/14 spurious "violations." This was a model/theory mismatch, not a refutation: the floor charges bits for *identifying* the load-bearing set, while that implementation let the oracle encode identity for free. The corrected model makes B a literal message length gating an encoder/decoder pair, restoring agreement. This is logged because the refute-first discipline is only honest if the near-misses are reported — and because it is a concrete instance of the deeper point: an information bound constrains *what a channel can carry*, and a simulation that smuggles information through an unpriced side channel will appear to beat it.

3.4 Key numbers

Quantity	Value
Floor $B^\star$ (catch $k = 2$ of $N = 60$, open $m = 8$)	5.98 bits
Floor, disjoint readers (catch $2k = 4$)	12.77 bits
Split penalty (disjoint − single)	6.78 bits
Ratio $\text{Floor}_{2k}/\text{Floor}_k$	2.13×
Super-additivity $\text{Floor}_{2k}/(2\,\text{Floor}_k)$	1.07
Oracle violations of the floor	0/16

4 What is now in hand for Paper 1

Three of Paper 1's four pillars are executed: A2 (the derived regret head, with the SDT direction corrected as a corollary), A1 (the comonotone characterization plus the super-additive split penalty), and A7 (the falsification figure with a passed test). The remaining pillar is B1 (the two-constraint rate–distortion frontier and the zoom-advantage theorem), which upgrades A7's single-budget floor into the full digest-bits vs. expected-opens trade. With B1, "The Price of a Summary" is a complete paper: a characterization, a frontier, a derived optimal policy, and data.

Reproducibility. The experiment is a single self-contained script (∼180 lines, numpy/scipy/matplotlib); the figure and all key numbers regenerate deterministically from seed `20260816`. It is the seed of the coordination benchmark the reviews recommend owning.